

Shubham Chaudhary

Senior Data Scientist | M.Tech Data Science & AI, IIT Patna | Bengaluru, India

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SUMMARY

Senior Data Scientist and AI Engineer with 3+ years of experience architecting and deploying production-grade intelligent systems across **Machine Learning, Deep Learning, NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG),** and **Agentic AI**. Proven track record of delivering measurable impact—**40–60% efficiency gains, 99.5% uptime, and 10,000+ concurrent users**—through scalable MLOps pipelines, multi-agent orchestration, and cloud-native deployments on **AWS, Azure, and GCP**. Adept at transforming complex business problems into intelligent automation that reduces operational overhead and accelerates delivery cycles.

SKILLS

- **Programming & Query:** Python, SQL, Bash/Shell Scripting
- **Machine Learning:** Supervised & Unsupervised Learning, Predictive Modeling, Classification, Regression, Clustering, Random Forest, XGBoost, CatBoost, Gradient Boosting, SVM, Ensemble Methods, Anomaly Detection, Time Series Forecasting, Feature Engineering, Hyperparameter Tuning, A/B Testing, Cross-Validation, Model Evaluation
- **Deep Learning & Computer Vision:** Neural Networks, CNNs, GANs, Transformers, Diffusion Models, Transfer Learning, Object Detection, PyTorch, TensorFlow, Keras, Lightning AI, Ultralytics YOLO, OpenCV
- **NLP & Large Language Models:** NLP, LLMs, BERT, GPT-4, GPT-4o, RAG, Prompt Engineering, LLM Fine-Tuning, Instruction Tuning, Semantic Search, Dense Embeddings, Hybrid Retrieval (Vector + BM25), Cross-Encoder Reranking, HyDE Query Expansion, Hard-Negative Mining, SentenceTransformers, Named Entity Recognition, Text Summarisation, Hugging Face Transformers
- **Agentic AI & Orchestration:** Agentic AI Systems, Multi-Agent Architecture, Model Context Protocol (MCP), Autonomous Agents, Workflow Automation, LangChain, LlamaIndex, AutoGen, CrewAI, Cognitive Architectures
- **AI Platforms:** OpenAI API (GPT-4, GPT-4o-mini), Anthropic Claude, Mistral AI, Groq, Ollama, Vertex AI, Amazon Bedrock, Weaviate
- **Vector Databases & Search:** Pinecone, ChromaDB, FAISS, Weaviate, Amazon OpenSearch Serverless, Neo4j
- **Cloud Platforms:** AWS (EC2, S3, RDS, SageMaker, ECS Fargate, Lambda, Bedrock, ALB), Azure (AKS, Azure ML, Azure DevOps, Terraform, Logic Apps), GCP (Vertex AI, BigQuery)
- **MLOps & DevOps:** MLflow, DVC, Databricks, Kubeflow, Jenkins, GitHub Actions, Azure DevOps, CI/CD Pipelines, Experiment Tracking, Model Registry, Model Monitoring, Automated Retraining, Prometheus, Grafana, Checkmarx One, SonarQube
- **Data Engineering:** Apache Kafka, Apache Spark, Hadoop, Hive, ETL Pipelines, Data Warehousing, Data Pipelines
- **Backend & API:** FastAPI, Flask, REST API, Microservices Architecture, Swagger UI, Postman, Webhooks
- **Containerisation & Orchestration:** Docker, Kubernetes
- **Databases:** MySQL, MongoDB, MariaDB, PostgreSQL, BigQuery
- **Data Visualisation:** Power BI, Tableau, Plotly, Matplotlib, Seaborn, Grafana
- **Statistical Methods:** Hypothesis Testing, A/B Testing, T-tests, ANOVA, Chi-Square, Bayesian Inference, Experimental Design, Statistical Inference
- **Methodologies:** Agile, Scrum, Cross-Functional Collaboration, Stakeholder Management, Technical Documentation

EXPERIENCE

Senior Data Scientist — Bristlecone, Bengaluru

Feb 2026 – Present

- **ScrumAI — Intelligent JIRA-to-Production Automation Platform:** Architected and shipped an AI-powered DevOps automation platform that eliminates **8+ manual coordination touchpoints per ticket** through a deterministic **9-state WorkflowStateMachine**, automating JIRA ticket parsing, Bitbucket branch creation, Jenkins CI/CD, SonarQube quality enforcement, Artemis AI (Code Buddy) PR review, and iterative fix-and-rebuild loops; reduced mean delivery cycle time by **40%** and enabled **50+ concurrent ticket workflows**.
- Engineered a **Model Context Protocol (MCP) multi-agent layer** coordinating 5 specialised agents — JiraAgent, BitbucketAgent, JenkinsAgent, SonarQubeAgent, and GPT-4o-mini CodingAgent — with intelligent state resumption from JIRA comments, **Checkmarx One REST** security scan integration, and auto-generated development prompts and fix guidance per ticket; reduced developer context-switching overhead by **60%**.
- Delivered full observability via **MLflow on Databricks**, 3-layer error handling including decorator classification, retry-backoff, and state-specific fallbacks, and a **FastAPI backend with sub-200ms response time**; resolved a critical Jenkins job-type routing bug using ground-truth `get_job_info` API classification with TTL caching and enum-based routing, improving pipeline trigger reliability by **70%**.

- **Project 1: Megatron RAG Chatbot System:** Engineered a production-grade **Retrieval-Augmented Generation (RAG)** chatbot ingesting **30GB+ of academic content**; transcribed **200+ tutor videos** via OpenAI Whisper, parsed 100+ books and research papers, and generated structured Q&A pairs via LLM; indexed content into **Pinecone** for low-latency semantic search and orchestrated the pipeline with **LangChain, LlamaIndex, and OpenAI GPT-4**, achieving a **60% improvement in query resolution efficiency**.
- Deployed on **AWS (EC2, S3)** with a FastAPI backend; integrated MLflow for experiment tracking and model versioning, and established real-time observability with **Prometheus & Grafana**; migration from ChromaDB to Pinecone reduced semantic retrieval latency by **45%** in production.
- **Project 2: Personalized Learning System — Multi-Agent MCP Architecture:** Designed an **MCP-based multi-agent adaptive learning system** coordinating Assessment Analyzer, Concept Mapper, and Content Recommender agents for data-driven study plan generation; integrated with the Megatron RAG chatbot for context-aware proactive recommendations, improving **course completion rates by 25%**.
- Deployed a containerised architecture on **AWS ECS Fargate** with Application Load Balancer (ALB) for elastic auto-scaling across **10,000+ concurrent users at 99.5% uptime**; implemented real-time monitoring of API latency and agent activity using **Prometheus & Grafana**.
- **Project 3: AI-Powered Talent Matching Engine (LLM + Hybrid Retrieval):** Built a production-grade ATS matching system using custom-trained semantic embeddings and **hybrid resume-JD retrieval (Vector Search + BM25)** to match candidates to job descriptions with explainable confidence-based ranking.
- Designed a multi-stage data pipeline with **GPT-4o synthetic resume generation, adversarial negatives, hard-negative mining, and leakage prevention**; trained a **SentenceTransformer bi-encoder** using supervised contrastive learning optimised for GPU-constrained environments.
- Deployed a scalable FastAPI service with **Cross-Encoder reranking, HyDE query expansion, automated PII redaction, and experience extraction**, improving **top-K matching precision by 35%** over keyword-only baselines.
- **Project 4: Industrial IoT Fault Detection & Auto-Retraining System:** Architected a containerised microservices pipeline on **Azure Kubernetes Service (AKS)** for real-time fault detection and automated model retraining using Docker, Apache Kafka, Apache Spark, MLflow, and Azure ML; implemented full CI/CD through **Azure DevOps & GitHub Actions** with automated retraining triggers and Prometheus/Grafana observability, maintaining **99.5% system uptime with zero manual retraining interventions**.

Associate Data Scientist — iNeuron.ai, Bengaluru

Jan 2023 – Oct 2023

- Designed a credit card fraud detection system using **XGBoost** with advanced feature engineering and Bayesian hyperparameter tuning, achieving an **F1-score of 80%** and reducing false positives by **35%**.
- Built a medical image analysis pipeline using **CNN with VGG16 transfer learning** on **50,000+ labelled images**, improving diagnostic classification accuracy by **20%** and reducing misdiagnosis rate by **15%**.

Data Science Intern — iNeuron.ai, Bengaluru

Jul 2022 – Jan 2023

- Implemented an unsupervised anomaly detection system for credit card transactions using **Isolation Forest and autoencoders** trained on **1M+ transaction records**, improving fraud detection accuracy by **25%** and reducing false alarm rate by **18%**.
- Developed **Mint News Summarizer** using Python, Selenium, BeautifulSoup, and NLP-based extractive/abstractive summarisation, scraping **500+ articles daily** and reducing manual reporting time from 4 hours to under 45 minutes (**70% reduction**).

PROJECTS

- **ScrumAI — JIRA-to-Production Automation:** Python, FastAPI, MCP, GPT-4o-mini, Jenkins, MLflow, Databricks, Checkmarx One; deterministic 9-state workflow, 5-agent orchestration, **40% cycle-time reduction** and **70% pipeline reliability improvement**.
- **Megatron RAG Chatbot — GenAI & RAG System:** LangChain, LlamaIndex, GPT-4, Pinecone, Whisper, AWS SageMaker, Mistral 7B; **30GB+** knowledge base with 2,000+ videos and 100+ books, achieving **60% improvement** in query resolution efficiency.
- **AI-Powered Talent Matching Engine:** SentenceTransformers, GPT-4o, Pinecone, BM25, Cross-Encoder, FastAPI; hybrid vector + keyword matching with hard-negative mining and contrastive learning, delivering **35% precision improvement** over keyword-based ATS.
- **Personalized Learning System — MCP Multi-Agent:** MCP, GPT-4, LangChain, AWS ECS Fargate, ALB, Prometheus; adaptive multi-agent learning platform with **25% improvement** in course completion, supporting **10,000+ concurrent users at 99.5% uptime**.
- **AI-Powered Industrial Monitoring — AIOps/MLOps:** AKS, Docker, Kafka, Spark, MLflow, Azure ML, Terraform, GitHub Actions; real-time fault detection and automated model retraining with **99.5% uptime** and zero manual retraining.

- **E-Commerce Data Warehouse & Analytics:** Apache Hive, Sqoop, Spark, HDFS, AWS S3, BigQuery; scalable ETL and data warehouse pipeline processing **10M+ product records**, improving data-processing efficiency by **30%**.

EDUCATION

M.Tech — Data Science and Artificial Intelligence	2026 – 2028
Indian Institute of Technology (IIT) Patna	<i>Pursuing</i>
B.Tech — Information Technology	2018 – 2022
Dr. A.P.J. Abdul Kalam Technical University	CGPA: 7.97 / 10.0
Senior School Certificate Examination (Class XII)	2015 – 2016
Jawahar Navodaya Vidyalaya, Basti, Uttar Pradesh	85.4%
Secondary School Examination (Class X)	2013 – 2014
Jawahar Navodaya Vidyalaya, Basti, Uttar Pradesh	CGPA: 10.0 / 10.0

CERTIFICATIONS

- **Full Stack Data Science Bootcamp** — iNeuron.ai
- **Deep Learning Specialization** — Coursera (Andrew Ng)
- **Google Data Analytics Professional Certificate** — Coursera
- **Docker for Professionals; Applied Machine Learning Algorithms** — LinkedIn Learning
- **Credit Card Fraud Detection (Internship Certificate)** — PW Skills

ACHIEVEMENTS

- **Star of the Month Award** — PW Skills
- **Special Initiative Award** — iNeuron.ai
- **On the Spot Award** — Knowledge sharing and team impact
- **Lifetime Member Award** — 4 years volunteering with “Koshish Ek Kadam Shiksha Ki Or”
- **Best Teacher Award; Best Academic Captain Award (2015)** — Jawahar Navodaya Vidyalaya